Encore–Sound Responsive Page Turner

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Motivation

Page turning can **disrupt a musician's rhythm and focus**, leading to **avoidable mistakes during performances**. Despite the rise of digital sheet music, this **issue persists**, highlighting **the need for a more seamless solution**.

Functionality

The device uses an **analog microphone** to capture and analyze audio in **real time**, identifying an end-of-page cue based on a predefined note pattern.

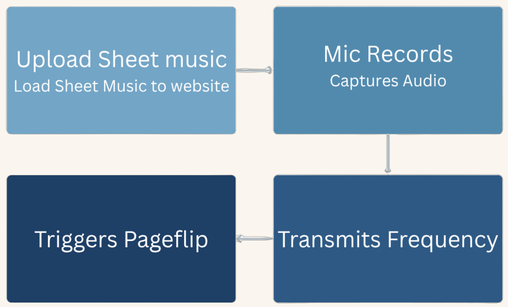
Once this cue is detected, the control system automatically triggers the page-turning mechanism.

A **DC motor** drives a **wheel** to **lift the page**, and a **servo arm completes the flip**, enabling **seamless**, **hands-free** operation during performance.

Software Overview

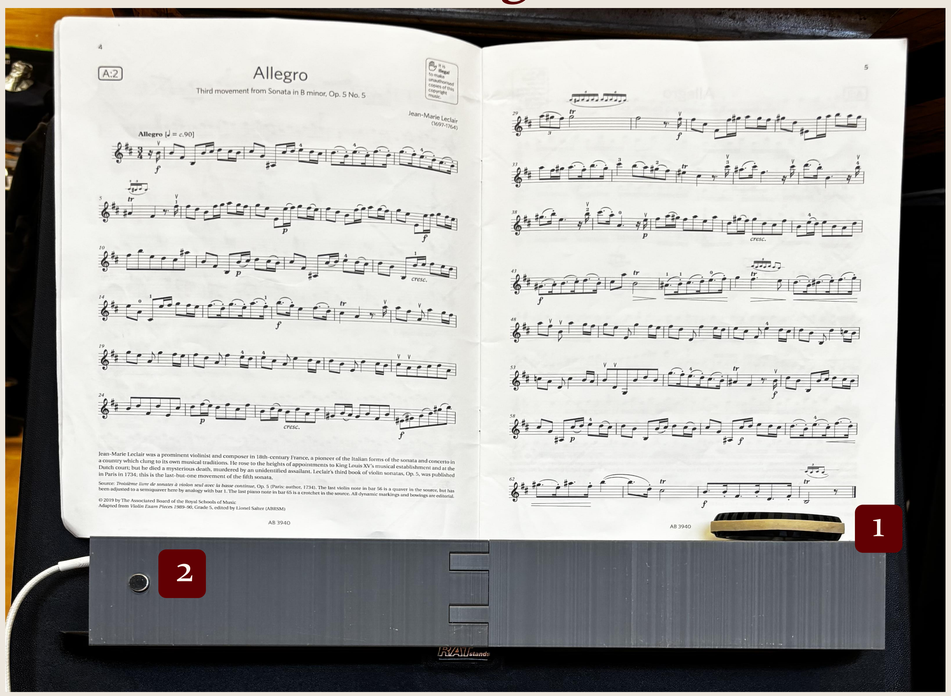
Encore converts a reference musical score into timed note events and uses them to track a live performance.

On the **ESP32**, microphone input is analyzed to estimate current pitch, and a **particle filter*** compares this live signal to the reference score to the estimate position in the music.



***Particle Filter:** a probabilistic model where particles represent states, evolve with noise, and are weighted and resampled to estimate the note.

Design





1 Wheel

- Starts to spin once the end of the page is detected
- Lifts up the page as it spins

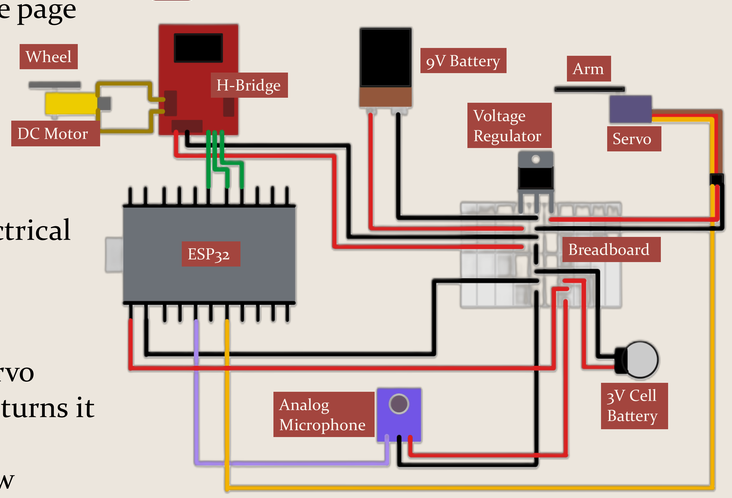
2 Analog Microphone

- MAX4466 analog microphone transmits frequencies to the electrical system

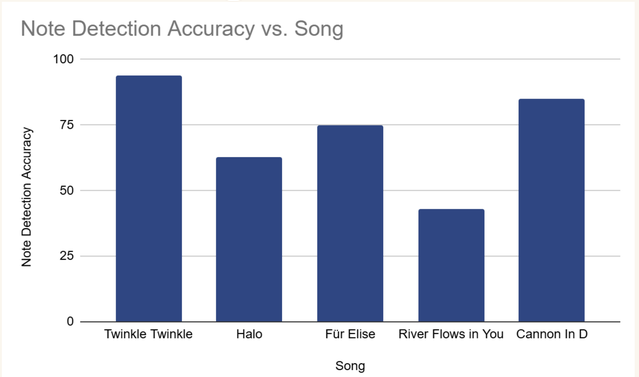
3 Arm

- 3D printed arm connected to servo
- Slides under the lifted page and turns it the rest of the way
- Large slot in the housing to allow movement

4 Electrical Circuit



Testing



Challenges

1. For the first iteration of the program, the accuracy of note detection decreased in noisier environments. To improve accuracy, we implemented a **particle filtering** function that filters out background audio.
2. Initial testing revealed that the music stand angle was <90°. Creating a **6° wedge** resolved the offset between the rotation axis and contact point of the wheel.
3. The second iteration revealed that flipping multiple pages reduced book thickness which reduced friction between the wheel and page. We added a **spring** to ensure books of **varying sizes** could be flipped.
4. While creating the final product we noticed a lack of space within our box for electronics. We removed the power source of the servo and added a **voltage regulator** to use a **common power source** for the servo and motor

Future Work

- Detect **chords** and **2 handed songs**
- Implement a **tension-adjustable** wheel or **sensor** to ensure consistent page lifting
- Adapt to a variety of **instruments**